DIS_TT · FINAL COMPARISON · 2026-11-03

TT-SVD vs DIS-TT on functions and their derivatives

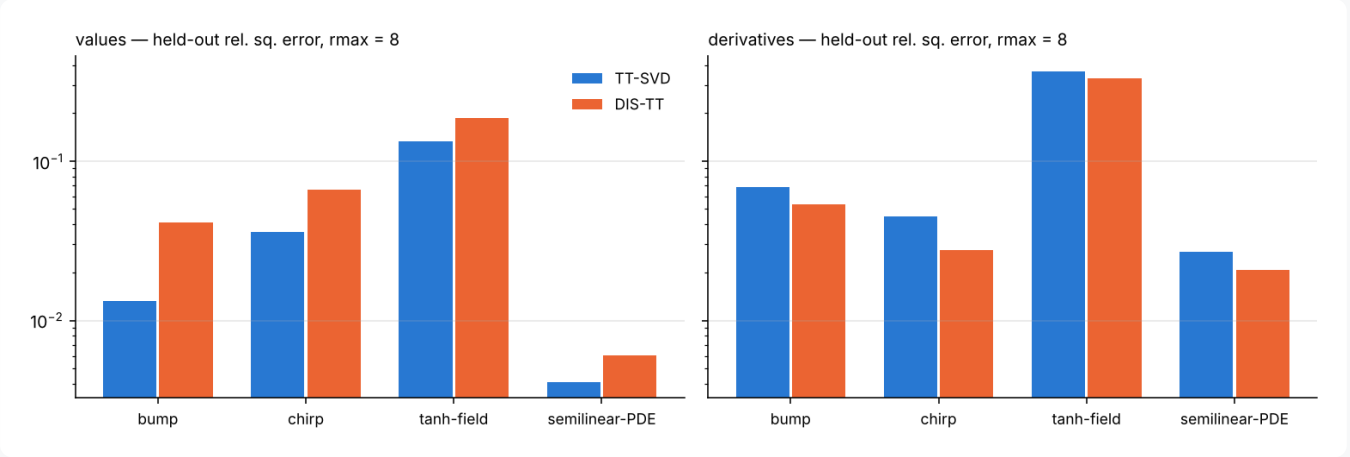
Two ways to build an r-column tensor-train basis for a family of functions u(x; m). TT-SVD (PCA-TT) compresses sampled values. DIS-TT compresses sampled Jacobians ∂u/∂m. Both run the same greedy sweep under the same rank cap. Only the snapshots differ.

TT-SVD (PCA-TT) DIS-TT

Setup

Parameters are m ~ N(0, I_p). The grids are QTT with 2^{10} points (2^8 for the PDE). Training uses N = 200 samples (100 for the PDE). Test sets have 200 samples (60 for the PDE).

The metric is the held-out relative squared projection error $\|(I - UU^T)A\|^2/\|A\|^2$. A is either the centred test values or the stacked test Jacobians. Analytic Jacobians come from the complex step. The PDE uses the whitened Jacobian DF·L from the repo's `SemilinearProblem`.



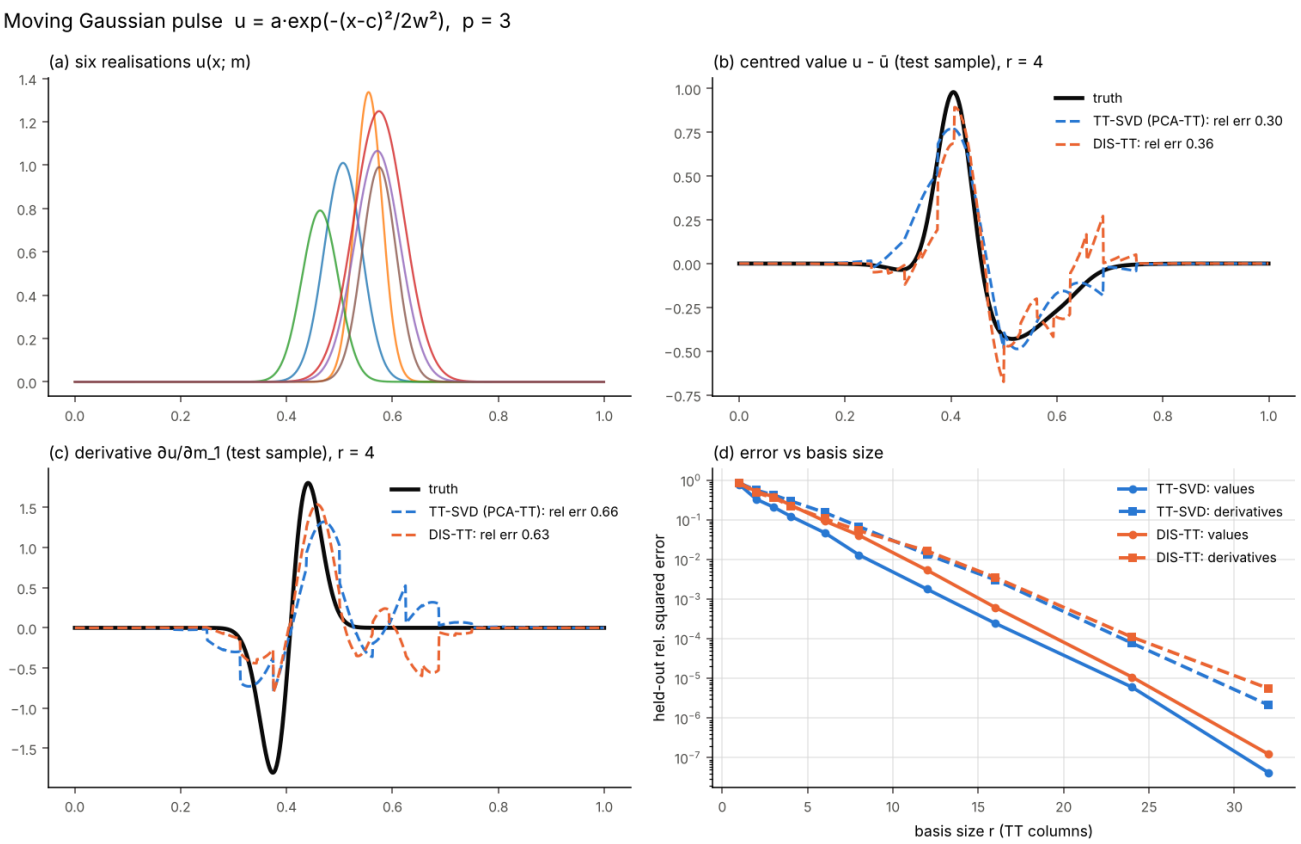
All four cases at rank cap 8, held-out relative squared error (log scale). TT-SVD wins on values in every case, and DIS-TT wins on derivatives in every case.

Matched basis size

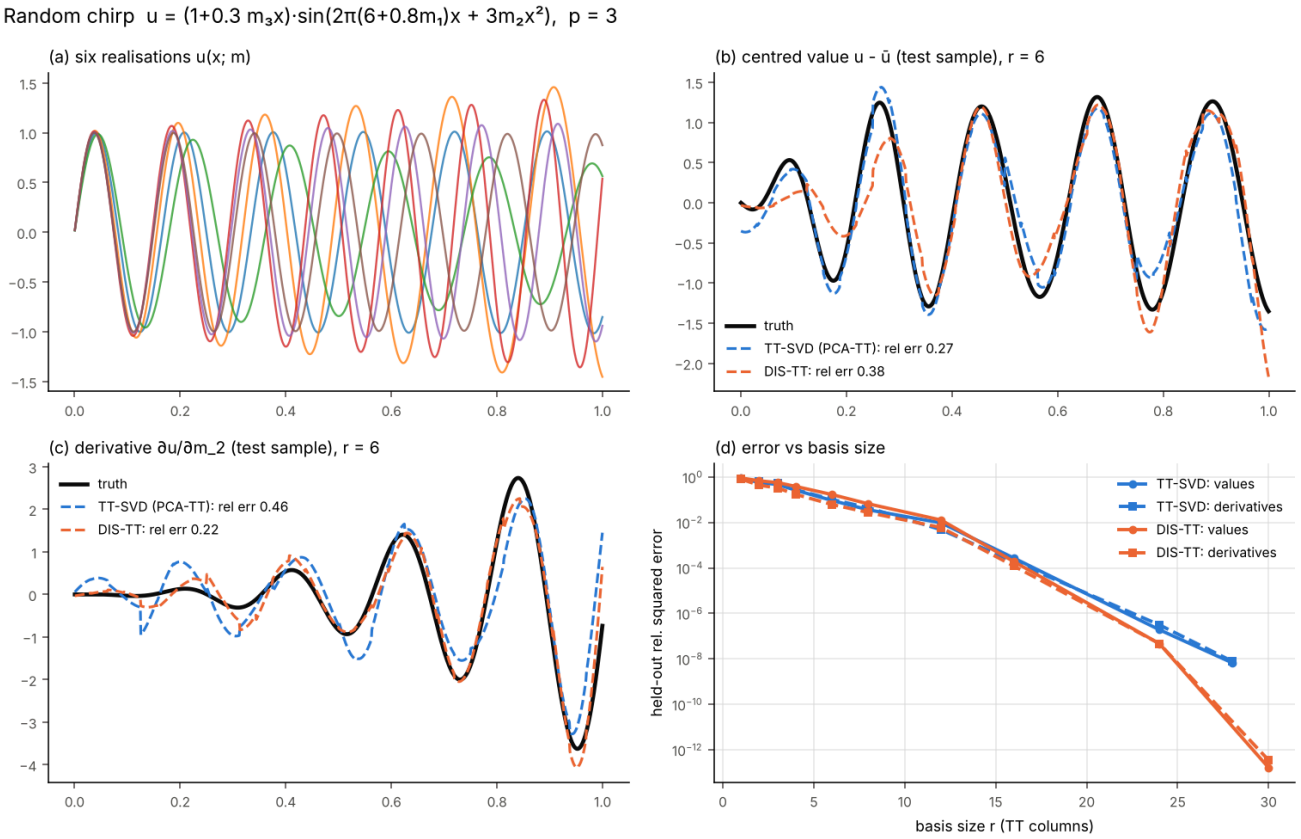
Case	r	Values · TT-SVD	Values · DIS-TT	Derivatives · TT-SVD	Derivatives · DIS-TT
bump	4	**1.25e-1**	2.38e-1	3.04e-1	**2.27e-1**
bump	8	**1.32e-2**	4.13e-2	6.86e-2	**5.38e-2**
bump	16	**2.46e-4**	6.21e-4	**3.08e-3**	3.54e-3
chirp	4	**2.64e-1**	3.88e-1	2.79e-1	**1.74e-1**
chirp	8	**3.61e-2**	6.62e-2	4.47e-2	**2.76e-2**
chirp	24	1.92e-7	**4.69e-8**	3.14e-7	**4.70e-8**
tanh-field	8	**1.32e-1**	1.85e-1	3.67e-1	**3.30e-1**
tanh-field	32	**4.94e-3**	6.96e-3	4.61e-2	**3.93e-2**
semilinear PDE	4	**3.06e-2**	3.77e-2	**1.17e-1**	1.58e-1
semilinear PDE	8	**4.09e-3**	6.05e-3	2.71e-2	**2.08e-2**
semilinear PDE	32	1.22e-5	**9.58e-6**	2.15e-5	**9.43e-6**

The shaded cell in each pair is the lower error. TT parameter counts are identical at each r, except near saturation (at most 7 % apart).

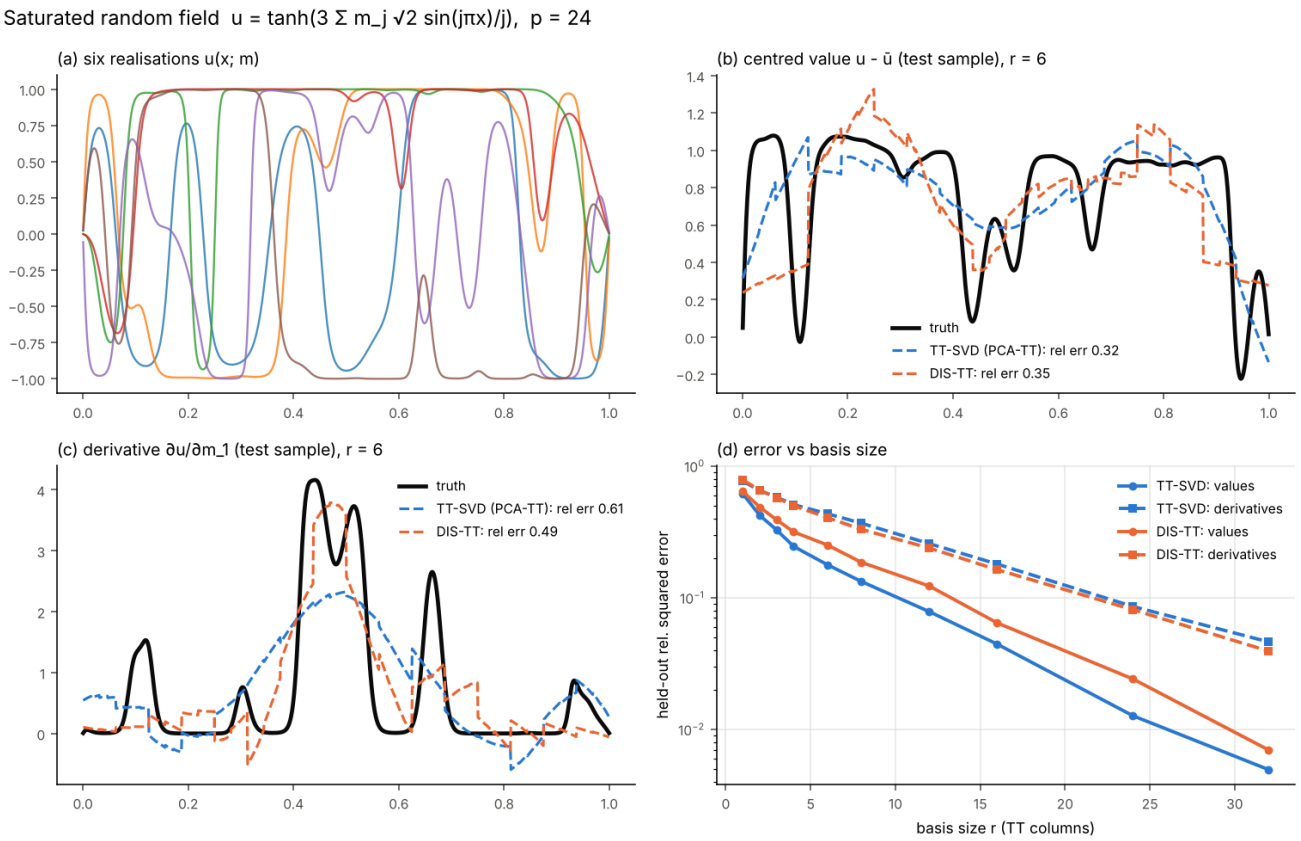
The four families



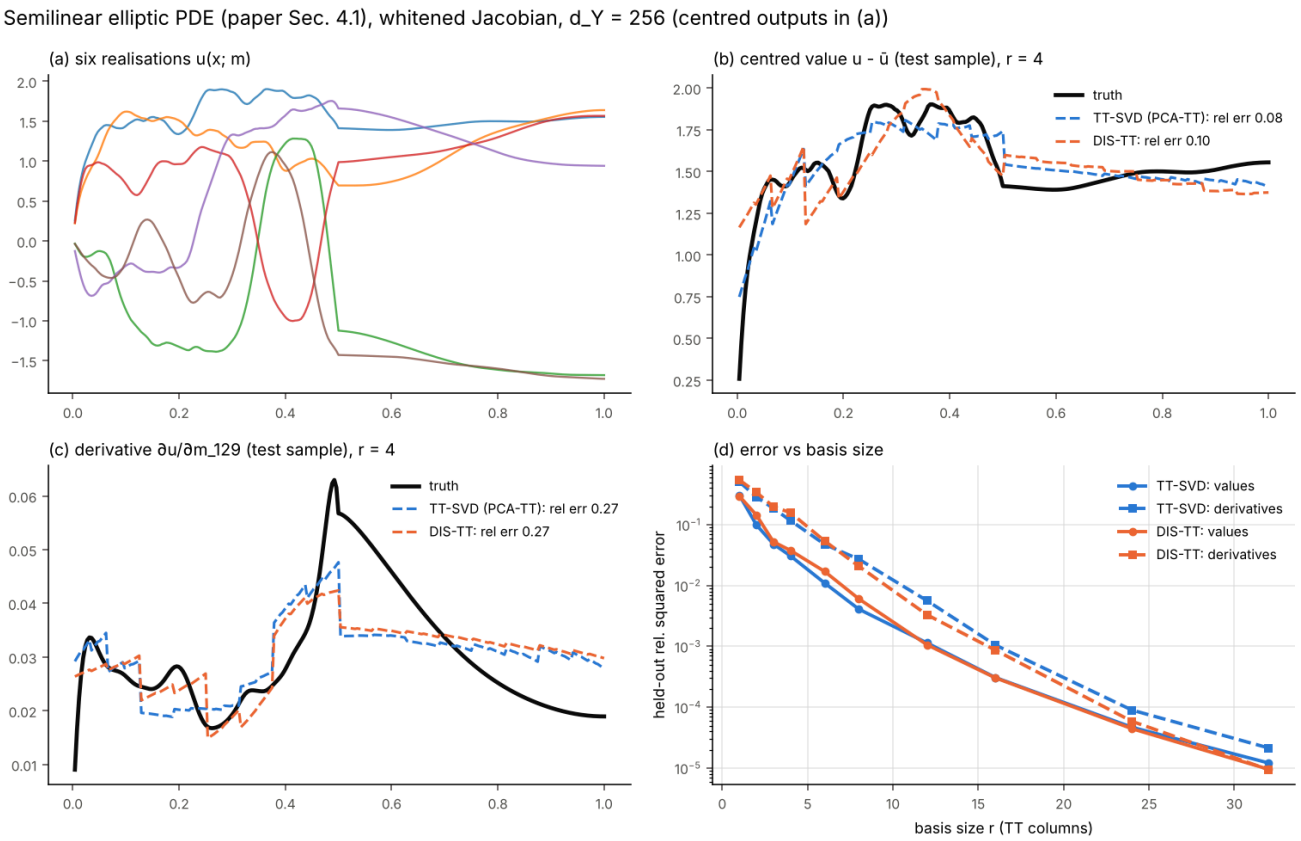
Moving pulse, a·exp(−(x−c)²/2w²) with random centre, width and amplitude (p = 3). Derivatives are pulse-shaped dipoles. DIS-TT tracks them better up to r = 8. From r = 12 on, TT-SVD is better on both values and derivatives.



Random chirp, $(1+0.3m_3x)\cdot\sin(2\pi(6+0.8m_1)x + 3m_2x^2)$ ($p = 3$). In panel (c) the TT-SVD projection of $\partial u/\partial m_2$ oscillates near $x = 0$, where the true derivative is flat. Above $r \approx 16$, DIS-TT gives the better basis for values too, reaching 1e-13 against about 1e-8.



Saturated random field, tanh(3 Σ m_j√2 sin(jπx)/j) with 24 parameters. The spectrum decays slowly for both methods, so rank is the bottleneck. DIS-TT keeps a small edge on derivatives, 2–15 % depending on r.



Semilinear elliptic PDE (paper Sec. 4.1, finite differences, $d_Y = 256$). The diffusion jump at x = 0.5 sets the structure. For r ≥ 8, DIS-TT is better on derivatives and roughly level on values, which matches the paper's prediction of output-side parity.

Findings

1. **Each basis wins the quantity it was built from.** At r = 8, DIS-TT cuts derivative error by 10–38 % in all four cases. TT-SVD has 1.4–3× lower value error.
2. **The gain is moderate, and it can reverse.** DIS-TT is at most about 1.6× better (chirp). On the pulse at r ≥ 12, and on the PDE at r = 4, TT-SVD is better on both quantities.
3. **Derivative snapshots can also make the better value basis.** For the chirp at high rank, the Jacobian columns span tangent directions that the value snapshots resolve poorly. DIS-TT is 4× better on values at r = 24.
4. **With many active parameters, rank is the limit.** On the 24-parameter tanh field neither basis reaches 1e-2 derivative error by r = 32.
5. **Compressing a single function is a different problem.** DIS-TT needs a parametric family with derivatives. A lone function u(x) can only be compressed by TT-SVD.

Caveats and reproduction

Each case uses one seed and one training set, with no error bars. The r ≤ 8 pattern holds in all four cases. The analytic cases use a whitened prior (L = I), so DIS-TT there is the uncentred TT-SVD of the Jacobian tensor.

To reproduce (about 40 s on CPU): `PYTHONPATH=src python experiments/plot_ttsvd_vs_dis.py`. This writes `docs/figures/`, including `ttsvd_vs_dis.json` with every rank's errors and parameter counts. The written report is `docs/TTSVD_vs_DIS_REPORT.md`. The full implementation status is in `FINAL_REPORT.md`.

APPENDIX

DIS-TT theory reference (condensed)

Full note, with derivations, complexity tables and caveats: https://claude.ai/artifact/Jjr4SEdweverCdqwKStCyn (section numbers below refer to it). Source paper: Luo, O'Leary-Roseberry, Chen, Ghattas, *Dimension reduction for derivative-informed operator learning*, arXiv:2504.08730 (2025).

Status of claims. The paper's results (DIS operators, Prop. 22 optimality, Lemma 39 sampling bounds, Thm. 10) are established. Every TT-specific statement here (Props. 1-2, the corollary, the input-side extension) was derived in the note, not taken from the paper, and is exactly what the test suite and experiments are meant to check.

Objects (note Sec. 2)

Symbol	Definition	Code
D_E F	DF(x) L, whitened Jacobian, C_X = L L^T	`ForwardModel.jvp` + `GaussianPrior.apply_sqrt`
H_Y	E[DF C_X DF^T] (output sensitivity)	`snapshots.output_dis_snapshots`
H_X^w	E[L^T DF^T DF L] (whitened input sensitivity)	`snapshots.input_dis_snapshots`
C_Y	Cov(F) (output PCA)	`snapshots.output_pca_snapshots`
Z	pre-scaled snapshots, Z Z^T = estimate of H	`SnapshotSet.Z`

Paper Prop. 22: E||(I-P)D_EF||_HS^2 = Tr((I-P) H_Y (I-P)), minimised by the top-r eigenvectors of H_Y with value sum_{i>r} lambda_i. Everything below applies this bond by bond.

TT projectors (note Sec. 3)

Left-orthogonal cores G_k (shape (r_{k-1}, n_k, r_k)); frames U_{<=k} = (U_{<=k-1} (x) I_{n_k}) G_k; projectors Pi_k = U_{<=k} U_{<=k}^T (x) I_{>k}, which are nested. With D the last bond:

```
||(I - Pi_D) A||^2 = sum_{k=1}^D ||(Pi_{k-1} - Pi_k) A||^2                    (T1)
E||(Pi_{k-1} - Pi_k) D_EF||^2 = Tr[(I - G_k G_k^T) H~_k]                      (T2)
H~_k = W_k^T Tr_{>k}(H_Y) W_k,   W_k = U_{<=k-1} (x) I_{n_k}                  (T3)
```

In snapshot form, Tr_{>k}(Z Z^T) is the Gram of the k-th left unfolding of the tensor (n_1, ..., n_d, T), so **DIS-TT = uncentred TT-SVD of the snapshot tensor with the snapshot index last**.

D = d gives an r-column block-TT basis (RBNO decoder). D = d-1 gives a TT subspace U_{<=d-1} (x) R^{n_d}.

Error statements (note Sec. 4)

```
Prop. 1:  max_k tail_{r_k}(H^(k)) <= min_TT err <= err_greedy = sum_k tail_{r_k}(H~_k)
                                                <= sum_k tail_{r_k}(H^(k)),     H^(k) = Tr_{>k} H
Exact on training data:  ||(I - Pi_D) Z||_F^2 == sum_k tails                    (tested to 1e-8)
Corollary (Gaussian Poincare): E||(I-Pi)(F - Fbar)||^2 <= E||(I-Pi) D_EF||^2
Prop. 2:  E_N err <= sum_k [ tail_{r_k}(H^(k)) + sqrt(2 r_k M_k / N) ],
          M_k = E||Tr_{>k}(D_EF D_EF^*) - H^(k)||_HS^2   (only the global branch is rigorous)
Rank rule: r_k minimal with tail_k <= eps^2 Tr(H_hat) / D   =>  cert <= eps^2
```

Input side: identical with H_X^w; V = L W is E-orthonormal; encoder W^T L^{-1} x. The H^1 ridge bound needs the paper's Assumption 9, which holds per direction and so sums over any E-orthonormal complement. Re-verify the paper's remaining proof steps before claiming it.

Known failure modes (note Sec. 8)

1. Prop. 5 (linear constraints preserved by output DIS) does **not** survive TT truncation.
2. Greedy and order-dependent: the factor D is real. ALS sweeps (`refine.py`) can only help.
3. The whitening factor L must be cheap in TT form. A wide L needs the pseudo-inverse and a ker(L) check (`input_side.py` docstring).
4. Per-bond derivative spectra may decay more slowly than PCA spectra. Untested; measure it.
5. M_k is unknown, so use bootstrap spreads (`bounds.bootstrap_tails`).
6. Non-identity mass matrices break TT structure unless (nearly) Kronecker.

Source: `docs/theory.md` in the repository.